SHEET 7 – **OPTIMIZATION FOR DATA SCIENCE** SS 2026
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Goal: We examine in this sheet logistic regression, a classification algorithm that predicts binary outcomes by modeling their probabilities.

Let be n I.I.D training datapoints $\{(x^{(i)}, y^{(i)})\}_{i=1}^{n}$, where each feature vector $x^{(i)} \in \mathbb{R}^d$, and each label $y^{(i)} \in \{0, 1\}$. For a single training datapoint (x, y), parameter vector $\theta \in \mathbb{R}^d$, and $\sigma : \mathbb{R} \to \mathbb{R}$ Sigmoid function, logistic regression assumes

$$P_\theta(Y = 1 \mid X = x) = \sigma(\theta^T x)$$

$$P_\theta(Y = 0 \mid X = x) = 1 - \sigma(\theta^T x)$$

Exercise 7.1

1. Determine the loss $\mathcal{J} : \mathbb{R}^d \to \mathbb{R}$ function equal to the negative log-likelihood function

$$\mathcal{J}(\theta) = -\log\left(\prod_{i=1}^{n} P_\theta(Y_i = y_i \mid x_i)\right).$$

2. Show that the loss function $\mathcal{J}$ is convex on $\mathbb{R}^d$.
3. Compute the gradient $\nabla\mathcal{J}(\theta)$.

Exercise 7.2

1. Consider the gradient descent iteration

$$\theta^{k+1} = \theta^k - \eta\,\nabla\mathcal{J}(\theta^k),$$

   where $\eta > 0$ is a fixed step size. Write a program that runs this iteration explicitly for logistic regression.

2. Consider the following dataset in $\mathbb{R}^2$:

$$x_1 = (1, 0), \quad y_1 = 1,$$
$$x_2 = (0, 1), \quad y_2 = 1,$$
$$x_3 = (-1, 0), \quad y_3 = 0,$$
$$x_4 = (0, -1), \quad y_4 =,$$
$$x_5 = (0.2, 0.2), \quad y_5 = 0.$$

   (a) Starting from $\theta^0 = 0$, run gradient descent numerically.

(b) Observe the behavior of $\|\theta^k\|$ and of $\mathcal{J}(\theta^k)$.

3. Consider the following dataset in $\mathbb{R}^2$:

$$x_1 = (2,2), \quad y_1 = 1,$$
$$x_2 = (3,2), \quad y_2 = 1,$$
$$x_3 = (-2,-2), \quad y_3 = 0,$$
$$x_4 = (-3,-2), \quad y_4 = 0.$$

(a) Starting from $\theta^0 = 0$, run gradient descent numerically.
(b) Observe the behavior of $\|\theta^k\|$ and of $\mathcal{J}(\theta^k)$.
(c) Compare your observations with previous question.

Exercise 7.3

1. Assume that the dataset is linearly separable. This means that there exists $u \in \mathbb{R}^d \setminus \{0\}$ such that

$$u^\top x_i > 0 \quad \text{whenever } y_i = 1, \qquad u^\top x_i < 0 \quad \text{whenever } y_i = 0.$$

(a) Show that

$$\inf_{\theta \in \mathbb{R}^d} \mathcal{J}(\theta) = 0.$$

(b) Does $\mathcal{J}$ have a minimizer? Why?

Exercise 7.4

1. Define the ℓ_2-regularized logistic loss

$$\mathcal{J}_\lambda(\theta) = \mathcal{J}(\theta) + \frac{\lambda}{2}\|\theta\|^2, \qquad \lambda > 0.$$

2. Show that $\mathcal{J}_\lambda$ is convex.

3. Show that $\mathcal{J}_\lambda$ is coercive.

4. Deduce that $\mathcal{J}_\lambda$ admits at least one minimizer.

5. Return to the dataset from Exercise 2, Q.3 and run gradient descent for $\mathcal{J}_\lambda$.

(a) Observe the behavior of $\|\theta^k\|$.
(b) Observe the behavior of $\mathcal{J}_\lambda(\theta^k)$.
(c) Compare this with the unregularized case. Conclude.